Mathematical Methods I

Differential 1-forms

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1. Covectors and 1-forms

A tangent vector describes a direction and a magnitude at a point. A 1-form evaluates that vector and produces a real number in a linear way.

Definition 1.1 (Covector and covector field)

Let $\Omega \subseteq \mathbb{R}^n$ be open. A **covector** at $p \in \Omega$ is a linear map

$$\omega_p : T_p\Omega \longrightarrow \mathbb{R}.$$

The set of these functionals is the **cotangent space** $T_p^*\Omega = (T_p\Omega)^*$, the dual of the tangent space. A **differential 1-form** ω assigns a covector ω_p to each point p; it is smooth when its coefficient functions (Section 3) are smooth.

Linearity refers to vectors based at the same point: for $u_p, v_p \in T_p\Omega$ and $a, b \in \mathbb{R}$,

$$\omega_p(au_p + bv_p) = a\omega_p(u_p) + b\omega_p(v_p).$$

Proposition 1.2 (Operations with 1-forms)

For 1-forms ϕ, ψ and a scalar function f, define

$$(\phi + \psi)_p(v_p) = \phi_p(v_p) + \psi_p(v_p),$$
$$(f\phi)_p(v_p) = f(p)\,\phi_p(v_p).$$

The factor f is evaluated at the base point of the vector; it is not differentiated.

Evaluation on a vector field

If X is a vector field, $\omega(X)$ is the scalar function

$$(\omega(X))(p) = \omega_p(X(p)).$$

In particular, $\omega_p(v_p)$ is a number, whereas $\omega(X)$ is a function of p.

Proposition 1.3 (Linearity over functions)

For vector fields X, Y and scalar functions f, g,

$$\omega(fX + gY) = f\,\omega(X) + g\,\omega(Y).$$

Thus, a 1-form transforms vector fields into functions by pointwise evaluation.

2. Coordinate differentials and the dual basis

Let $x^1, \dots, x^n$ be the Cartesian coordinate functions. We write

$$U_j = \partial_j = \frac{\partial}{\partial x^j}, \qquad v_p = \sum_{j=1}^{n} v^j \, \partial_j|_p.$$

The superscripts in x^i and v^i are indices, not powers.

Definition 2.1 (Differential of a function)

For a differentiable function $f : \Omega \to \mathbb{R}$, its differential at p is

$$(df)_p(v_p) = v_p[f].$$

The differentiability of f guarantees that this expression is linear in v_p. If f is smooth, df is a smooth 1-form. Applied to $f = x^i$,

$$(dx^i)_p(v_p) = v_p[x^i] = v^i.$$

The coordinate 1-form dx^i selects the ith component of the vector.

Proposition 2.2 (Duality of the bases)

At each point, $\{(dx^1)_p, \dots, (dx^n)_p\}$ is the dual basis of $\{\partial_1|_p, \dots, \partial_n|_p\}$. Indeed,

$$(dx^i)_p(\partial_j|_p) = \partial_j|_p[x^i] = \frac{\partial x^i}{\partial x^j}(p) = \delta^i_j, \qquad \delta^i_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

As an identity between functions, we write $dx^i(U_j) = \delta^i_j$.

Note that $dx^i(v_p) = v^i$ reads off a component of the vector: the output depends only on v_p, not on a choice of metric.

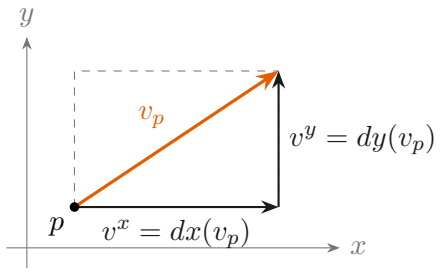


Figure 1: The forms dx and dy return the components of the vector based at p. The horizontal and vertical arrows represent vector components, not the 1-forms themselves.

3. Expression and evaluation of a 1-form

Proposition 3.1 (Coordinate expression)

Every 1-form can be uniquely expressed as

$$\omega = \sum_{i=1}^{n} a_i \, dx^i,$$

where the a_i are scalar functions. Its coefficients are recovered by evaluating it on the tangent basis:

$$a_i = \omega(U_i).$$

Proof

At a point p, let $v_p = \sum_j v^j \partial_j|_p$. By linearity and duality,

$$\begin{aligned}\omega_p(v_p) &= \left(\sum_i a_i(p)(dx^i)_p\right)\left(\sum_j v^j \partial_j|_p\right) \\ &= \sum_i \sum_j a_i(p) v^j \, (dx^i)_p(\partial_j|_p) \\ &= \sum_i \sum_j a_i(p) v^j \delta^i_j = \sum_i a_i(p) v^i.\end{aligned}$$

In particular,

$$\omega(U_j) = \sum_i a_i \, dx^i(U_j) = \sum_i a_i \delta^i_j = a_j.$$

Conversely, defining $a_i = \omega(U_i)$, the form $\sum_i a_i dx^i$ agrees with ω on every basis vector and, by linearity, on every vector. This proves the existence and uniqueness of the expression.

Proposition 3.2 (Evaluation formula)

If $X = \sum_i X^i \partial_i$, then

$$\boxed{\omega(X) = \sum_i a_i X^i.}$$

Both sides are functions; evaluation at p gives $\omega_p(X(p))$.

Example 3.3 (A function and its value at a point)

In $\mathbb{R}^2$, let $\omega = y\,dx + 2x\,dy$ and $X = x\partial_x - \partial_y$. Then

$$\omega(X) = xy - 2x.$$

For $p = (1, 3)$, $\omega_p = 3(dx)_p + 2(dy)_p$ and $X(p) = (1, -1)_p$, so $\omega_p(X(p)) = 3 - 2 = 1$. Moreover, $\omega(\partial_x) = y$ and $\omega(\partial_y) = 2x$ recover the coefficients.

4. Geometric picture: a covector is a stack of lines

A vector is an arrow. A covector has an equally concrete picture, but it is not an arrow: it is a family of equally spaced parallel lines (planes, in $\mathbb{R}^3$).

Picture 4.1 (Covector as a stack)

Given a nonzero covector ω_p, draw the lines

$$\omega_p(v) = k, \qquad k \in \mathbb{Z},$$

in the tangent space. The line $k = 0$ is the kernel: the vectors that ω_p sends to zero. Then

$$\omega_p(v_p) = \text{number of lines crossed by the arrow } v_p$$

(counted with sign, and fractionally if the arrow stops between two lines). Doubling v_p doubles the count and adding arrows adds the counts; this is exactly linearity. The more tightly packed the lines, the larger ω_p.

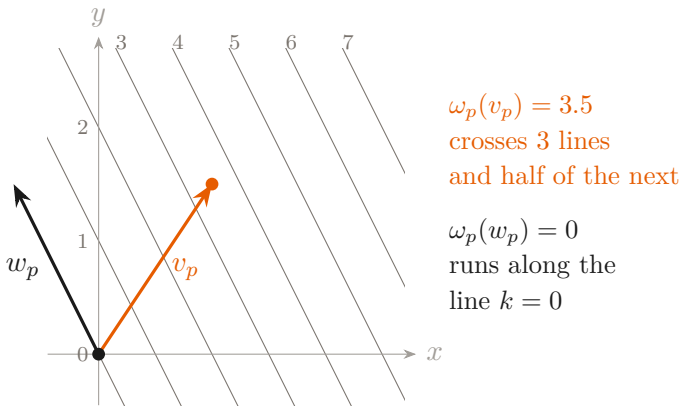


Figure 2: The covector $\omega_p = 2\,dx + dy$ drawn as the lines $2x + y = k$. The arrow $v_p = (1, 1.5)$ crosses lines $1, 2, 3$ and stops halfway to 4, so $\omega_p(v_p) = 3.5$. The arrow $w_p = (-0.5, 1)$ lies on the kernel and crosses nothing. (The picture lives in the tangent space at p; the origin stands for p.)

Example 4.2 (Reading the picture)

For $\omega_p = 2\,dx + dy$ and $v_p = (1, 1.5)$, the formula of Proposition 3.2 gives $2(1) + 1(1.5) = 3.5$, matching the count in Figure 2. For $w_p = (-0.5, 1)$ we get $2(-0.5) + 1 = 0$. Linearity is visible too: $\omega_p(v_p + w_p) = 3.5 + 0$, and $v_p + w_p = (0.5, 2.5)$ gives $1 + 2.5 = 3.5$.

The differential df and level lines

For df_p the lines are the level lines of f near p, drawn at unit steps of f. Then $(df)_p(v_p)$ is the number of level lines crossed by v_p, that is, the (first-order) change of f when we move along v_p. A vector tangent to the level set of f crosses no lines and gives 0.

The *gradient* is the arrow perpendicular to the lines, pointing toward increasing k, whose length

is the density of the lines. Notice that "perpendicular" and "length" need a metric, whereas the stack of lines does not. This is why df is the more basic object (see the last box of Section 9).

Looking ahead: integration

This picture is what makes 1-forms natural objects to integrate. Along a curve γ, the integral $\int_\gamma \omega$ adds up the lines crossed by the velocity of γ at each instant. If $\omega = df$, the total count is just the net number of level lines crossed, $f(\text{end}) - f(\text{start})$, which is zero on a closed loop.

5. The differential df and exact forms

Theorem 5.1 (Coordinate expression of the differential)

For a differentiable function $f : \Omega \to \mathbb{R}$,

$$\boxed{df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}\, dx^i.}$$

A 1-form that can be written as df throughout its domain is called **exact**. The function f is a potential for that form.

Proof by evaluation

For any $v_p = \sum_j v^j \partial_j|_p$, the definition of the differential gives

$$(df)_p(v_p) = v_p[f] = \sum_i v^i \frac{\partial f}{\partial x^i}(p).$$

On the other hand, using duality,

$$\begin{aligned}\left(\sum_i \frac{\partial f}{\partial x^i}(p)(dx^i)_p\right)(v_p) &= \sum_i \sum_j \frac{\partial f}{\partial x^i}(p) v^j \delta^i_j \\ &= \sum_i \frac{\partial f}{\partial x^i}(p) v^i.\end{aligned}$$

The two forms agree on all tangent vectors and are therefore equal. Equivalently, their coefficients are obtained as

$$df(U_i) = U_i[f] = \frac{\partial f}{\partial x^i}.$$

Example 5.2 (Computing df)

If $f(x, y, z) = x^2 y + z^2$, then

$$df = 2xy\, dx + x^2\, dy + 2z\, dz.$$

For $p = (1, 2, -1)$ and $v_p = (2, -1, 3)_p$,

$$(df)_p(v_p) = 4(2) + 1(-1) - 2(3) = 1.$$

This gives the directional derivative of f along the vector v_p.

Differential, vector field, and curve

For any vector field X, $df(X) = X[f]$. If γ is a differentiable curve contained in Ω, the chain rule states that

$$(df)_{\gamma(t)}(\gamma'(t)) = \frac{d}{dt}(f(\gamma(t))).$$

Evaluating df on the velocity measures how f changes along the curve.

6. The operator d: products and compositions

The operator d transforms a scalar function f into the 1-form df. The usual differentiation rules thus become equalities between 1-forms. We assume that the functions are differentiable on their domains.

Proposition 6.1 (Linearity and the product rule)

For $f, g : \Omega \to \mathbb{R}$ and constants $a, b \in \mathbb{R}$,

$$d(af + bg) = a\,df + b\,dg, \qquad d(c) = 0 \quad (c \text{ constant}).$$

The **Leibniz rule** states that

$$\boxed{d(fg) = f\,dg + g\,df.}$$

The factors f and g multiply 1-forms. At a point p, this equality means

$$\big(d(fg)\big)_p = f(p)(dg)_p + g(p)(df)_p.$$

Why the product rule holds

For any tangent vector v_p, we apply the definition of the differential and the product rule for directional derivatives:

$$\begin{aligned}\big(d(fg)\big)_p(v_p) &= v_p[fg] \\ &= f(p)\,v_p[g] + g(p)\,v_p[f] \\ &= f(p)(dg)_p(v_p) + g(p)(df)_p(v_p).\end{aligned}$$

Since both 1-forms give the same result on every v_p, they are equal. Linearity is justified in the same way.

Example 6.2 (Product of two functions)

Let $f(x,y) = x^2$ and $g(x,y) = y$. First, we compute

$$df = 2x\,dx, \qquad dg = dy.$$

Then we apply the rule:

$$\begin{aligned} d(x^2y) &= x^2\,d(y) + y\,d(x^2) \\ &= x^2\,dy + 2xy\,dx. \end{aligned}$$

This gives the term corresponding to x^2y in Example 5.2. Since $d(z^2) = 2z\,dz$, linearity gives

$$d(x^2y + z^2) = 2xy\,dx + x^2\,dy + 2z\,dz.$$

In $f\,dg$, the function multiplies the form: $(f\,dg)_p(v_p) = f(p)(dg)_p(v_p)$. Two differentials are not multiplied together here; $d(fg)$ is a 1-form.

Proposition 6.3 (Chain rule)

Let $f : \Omega \to \mathbb{R}$ and $h : A \to \mathbb{R}$, where $A \subseteq \mathbb{R}$ is open and $f(\Omega) \subseteq A$. Then

$$\boxed{d(h \circ f) = (h' \circ f)\,df = h'(f)\,df.}$$

The outer function h is differentiated, its derivative is evaluated at f, and the result is multiplied by the differential of the inner function.

Proof by evaluation

For every tangent vector v_p, the ordinary chain rule gives

$$\big(d(h \circ f)\big)_p(v_p) = v_p[h \circ f] = h'(f(p))\,v_p[f] = h'(f(p))\,(df)_p(v_p).$$

This proves the equality of the 1-forms.

Example 6.4 (A composition step by step)

For $F(x,y) = \sin(x^2y)$, the inner function is $f(x,y) = x^2y$ and the outer function is $h(u) = \sin u$. We have

$$h'(u) = \cos u, \qquad df = 2xy\,dx + x^2\,dy.$$

Therefore,

$$dF = \cos(x^2y)(2xy\,dx + x^2\,dy).$$

The factor $\cos(x^2y)$ multiplies the entire 1-form df.

Composition with several inner functions

If $F = H(f_1, \ldots, f_m)$, where H is differentiable on an open set containing the image of $(f_1, \ldots, f_m)$,

then

$$\boxed{dF = \sum_{a=1}^{m} \frac{\partial H}{\partial u^a}(f_1, \ldots, f_m)\, df_a.}$$

For example, for $F = e^{xy} + z^2$, take $H(u, v) = e^u + v^2$, $f_1 = xy$, and $f_2 = z$. Thus,

$$dF = e^{xy}\, d(xy) + 2z\, dz = e^{xy}(y\, dx + x\, dy) + 2z\, dz.$$

Useful consequences

$$d(f^k) = k f^{k-1} df \quad (k = 1, 2, \ldots), \qquad d(e^f) = e^f df.$$

Combining the product and composition rules, where $g \neq 0$,

$$d\left(\frac{f}{g}\right) = \frac{1}{g}\, df - \frac{f}{g^2}\, dg = \frac{g\, df - f\, dg}{g^2}.$$

7. Indices and coordinate changes

The components of a vector and a 1-form change with the coordinates, although the geometric object and its evaluation remain the same. From here on we write ω_i for the coefficients a_i of Section 3.

Convention 7.1 (Einstein summation)

An index repeated once as a superscript and once as a subscript indicates summation:

$$X = X^i \partial_i, \qquad \omega = \omega_i\, dx^i, \qquad \omega(X) = \omega_i X^i.$$

The repeated index is a **dummy index**: it may be renamed without changing the expression. Free indices must match on both sides of an equality.

The vector components X^i are **contravariant**; the components ω_i of a 1-form are **covariant**. The tangent basis ∂_i and the cotangent basis dx^i transform so as to compensate for the change in their respective components.

A fixed vector, different coordinate descriptions

Think of a tangent vector as a fixed arrow at a point p, for example the velocity of a particle at one instant. The arrow is the geometric object; its components depend on the coordinates used to describe it.

Let $x = (x^1, \ldots, x^n)$ and $u = (u^1, \ldots, u^n)$ be two local coordinate systems, with invertible change of coordinates $x = x(u)$. Set

$$J^i{}_a = \frac{\partial x^i}{\partial u^a}, \qquad (J^{-1})^a{}_i = \frac{\partial u^a}{\partial x^i}.$$

Let $\gamma(t)$ be a curve with $\gamma(0) = p$ and $v_p = \gamma'(0)$. In coordinates x and u, the same motion has

components

$$v_x^i = \left.\frac{d(x^i \circ \gamma)}{dt}\right|_{t=0}, \qquad v_u^a = \left.\frac{d(u^a \circ \gamma)}{dt}\right|_{t=0}.$$

Applying the chain rule to $u^a = u^a(x)$ gives

$$\boxed{v_u^a = \frac{\partial u^a}{\partial x^i}(p)\, v_x^i,} \qquad \text{that is,} \qquad \boldsymbol{v}_u = J(p)^{-1}\boldsymbol{v}_x.$$

The transformation law for vector components is simply the chain rule for the velocity of the same curve; the Jacobian converts components locally at p.

Proposition 7.2 (Transformation of bases and components)

By the chain rule,

$$\partial_{u^a} = \frac{\partial x^i}{\partial u^a}\partial_{x^i}, \qquad du^a = \frac{\partial u^a}{\partial x^i}dx^i,$$
$$X_u^a = \frac{\partial u^a}{\partial x^i}X_x^i, \qquad \omega_{u,a} = \frac{\partial x^i}{\partial u^a}\omega_{x,i}.$$

Writing each list as a column, these identities become

$$\boldsymbol{\partial}_u = J^T\boldsymbol{\partial}_x, \qquad d\boldsymbol{u} = J^{-1}d\boldsymbol{x}, \qquad \boldsymbol{X}_u = J^{-1}\boldsymbol{X}_x, \qquad \boldsymbol{\omega}_u = J^T\boldsymbol{\omega}_x.$$

Invariance of evaluation

Contracting a covariant index with a contravariant one gives

$$\omega_{u,a}X_u^a = \frac{\partial x^i}{\partial u^a}\omega_{x,i}\frac{\partial u^a}{\partial x^j}X_x^j = \delta_j^i\omega_{x,i}X_x^j = \omega_{x,i}X_x^i.$$

An index repeated twice as a superscript or twice as a subscript does not represent a contraction under this convention. For example, the inner product is written $g_{ij}A^iB^j$; in an orthonormal Cartesian basis it can be simplified to $\sum_i A^iB^i$, because there $g_{ij} = \delta_{ij}$.

8. Example: polar coordinates

Consider $x = r\cos\theta$, $y = r\sin\theta$, with $r > 0$ and a smooth local choice of the angle. The origin lies outside this chart.

Proposition 8.1 (Jacobian and inverse matrix)

With $J = \partial(x, y)/\partial(r, \theta)$,

$$J = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \qquad J^{-1} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta/r & \cos\theta/r \end{pmatrix}.$$

We have $\det J = r \neq 0$.

Tangent basis and vector components

$$\begin{aligned}
\partial_r &= \cos\theta\,\partial_x + \sin\theta\,\partial_y,\\
\partial_\theta &= -r\sin\theta\,\partial_x + r\cos\theta\,\partial_y,\\
V^r &= \cos\theta\,V^x + \sin\theta\,V^y,\\
V^\theta &= -\frac{\sin\theta}{r}V^x + \frac{\cos\theta}{r}V^y.
\end{aligned}$$

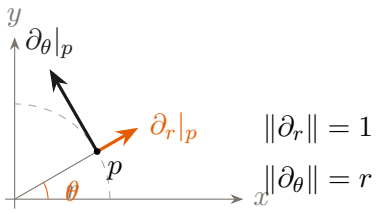


Figure 3: Polar coordinate basis, illustrated at $r = 2$. The vector ∂_θ is tangent to the circle and has length r; it is not generally a unit vector.

Cotangent basis and 1-form components

$$\begin{aligned}
dr &= \cos\theta\,dx + \sin\theta\,dy,\\
d\theta &= -\frac{\sin\theta}{r}\,dx + \frac{\cos\theta}{r}\,dy,\\
\omega_r &= \cos\theta\,\omega_x + \sin\theta\,\omega_y,\\
\omega_\theta &= -r\sin\theta\,\omega_x + r\cos\theta\,\omega_y.
\end{aligned}$$

These formulas verify that $dr(\partial_r) = d\theta(\partial_\theta) = 1$ and $dr(\partial_\theta) = d\theta(\partial_r) = 0$. With the columns and Jacobian defined above, the correct relation is

$$\begin{pmatrix} dr \\ d\theta \end{pmatrix} = J^{-1}\begin{pmatrix} dx \\ dy \end{pmatrix}.$$

9. Verification in both coordinate systems

Example 9.1 (A 1-form and a vector)

Consider

$$\omega = x\,dx + y\,dy, \qquad V = \partial_x.$$

In Cartesian coordinates, $\omega(V) = x$. In polar coordinates, the components of the form are

$$\omega_r = x\cos\theta + y\sin\theta = r,$$
$$\omega_\theta = -rx\sin\theta + ry\cos\theta = 0.$$

Therefore, $\omega = r\,dr$. The components of the same vector are

$$V^r = \cos\theta, \qquad V^\theta = -\frac{\sin\theta}{r}.$$

Evaluation in the polar basis gives

$$\omega(V) = \omega_r V^r + \omega_\theta V^\theta = r\cos\theta = x.$$

The results agree: evaluation does not depend on the coordinates.

The same form as an exact differential

Define

$$f(x,y) = \frac{x^2+y^2}{2}.$$

In Cartesian coordinates, $df = x\,dx + y\,dy$. In polar coordinates, the same function is expressed as $f = r^2/2$, so

$$df = r\,dr.$$

The two expressions describe the same 1-form. The potential f exists on all of $\mathbb{R}^2$, although polar coordinates are used only locally away from the origin.

Differential and gradient

The differential df is a covector. In Euclidean space, the inner product relates it to the gradient vector:

$$(df)_p(v_p) = \nabla f(p)\cdot v.$$

This identification uses the metric; the definition of df as a linear functional does not require it.

10. Connection with classical mechanics

Generalized coordinates and generalized velocities

In classical mechanics, a configuration is a point of configuration space. Generalized coordinates $q^1, \dots, q^m$ are local coordinates on that space; they may be lengths, angles, or other independent parameters. A motion is a curve $q(t)$, and its tangent vector is

$$\dot\gamma(t) = \dot q^a(t)\, \partial_{q^a}\big|_{\gamma(t)}, \qquad dq^a(\dot\gamma) = \dot q^a.$$

The numbers $\dot q^a$ are the **generalized velocities**. They are coordinate components, so they need not be speeds measured along unit vectors. For example, $\dot\theta$ is an angular velocity, whereas $r\dot\theta$ is a tangential speed.

For a particle with a time-independent position map $\mathbf{r} = \mathbf{r}(q)$, the physical velocity is

$$\boxed{\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{\partial \mathbf{r}}{\partial q^a}\, \dot q^a.}$$

For an unconstrained particle with $m = n$ and an invertible coordinate map, this is precisely $\boldsymbol{v}_x = J\dot{\boldsymbol{q}}$, with $J = \partial x/\partial q$, or $\dot{\boldsymbol{q}} = J^{-1}\boldsymbol{v}_x$ (Section 7). With constraints, m may be smaller than the number of ambient Cartesian coordinates: the chain rule still holds, but the position Jacobian is then rectangular and has no ordinary inverse. For explicitly time-dependent maps $\mathbf{r}(q,t)$, the velocity also includes $\partial\mathbf{r}/\partial t$.

Forces as 1-forms: virtual work and power

For a force $\mathbf{F}$ on a particle, the Euclidean inner product defines a work 1-form on configuration space:

$$\eta = \mathbf{F}\cdot d\mathbf{r} = Q_a\, dq^a, \qquad \boxed{Q_a = \mathbf{F}\cdot\frac{\partial \mathbf{r}}{\partial q^a}.}$$

The numbers Q_a are the **generalized forces**. Evaluating η on a displacement δq gives the **virtual work** $\delta W = Q_a\,\delta q^a$, and evaluating it on the velocity gives the **power**,

$$\eta(\dot\gamma) = Q_a \dot q^a = \mathbf{F}\cdot\mathbf{v}.$$

The last equality follows from the boxed velocity formula. Since the Q_a multiply the differentials dq^a, they are *covariant* components and transform as in Proposition 7.2: $Q_{u,a} = \frac{\partial q^b}{\partial u^a} Q_{q,b}$. The generalized velocities, in contrast, are contravariant.

Example 10.2 (Generalized forces in polar coordinates)

Let $\mathbf{F} = F_r\,\mathbf{e}_r + F_\theta\,\mathbf{e}_\theta$, where $\mathbf{e}_r = \partial_r$ and $\mathbf{e}_\theta = \partial_\theta/r$ are the unit vectors of Section 8. With $\mathbf{r} = (r\cos\theta, r\sin\theta)$,

$$Q_r = \mathbf{F}\cdot\partial_r = F_r, \qquad Q_\theta = \mathbf{F}\cdot\partial_\theta = r\,F_\theta, \qquad \eta = F_r\,dr + r\,F_\theta\,d\theta.$$

Q_θ is a torque (force times lever arm), not a force: the generalized forces take the units that make $Q_a\,dq^a$ a work. This agrees with the rule $\omega_\theta = r(\cdots)$ of Section 8.

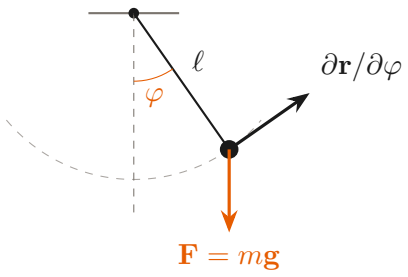


Figure 4: A pendulum has one generalized coordinate φ in a two-dimensional space. The vector $\partial\mathbf{r}/\partial\varphi$ (length ℓ, drawn shorter) is tangent to the circle of motion; only the component of $\mathbf{F}$ along it contributes to Q_φ.

Example 10.3 (Pendulum: a constraint and an exact form)

A bob of mass m hangs from a rod of length ℓ. With y pointing up, the position is $\mathbf{r}(\varphi) = (\ell\sin\varphi,\, -\ell\cos\varphi)$, so $m = 1$ while the ambient space has $n = 2$: the Jacobian $\partial\mathbf{r}/\partial\varphi = \ell(\cos\varphi, \sin\varphi)$ is a 2×1 column and has no inverse. For gravity, $\mathbf{F} = (0, -mg)$,

$$Q_\varphi = \mathbf{F}\cdot\frac{\partial\mathbf{r}}{\partial\varphi} = -mg\ell\sin\varphi, \qquad \eta = -mg\ell\sin\varphi\, d\varphi.$$

This is the restoring torque. Moreover η is exact: with $V(\varphi) = -mg\ell\cos\varphi$ we have $dV = mg\ell\sin\varphi\, d\varphi$, so

$$\eta = -dV.$$

A force whose work 1-form is $-dV$ is called *conservative*: by the remark at the end of Section 4, the work done around any closed path in configuration space is 0.